1. In micro-programmed approach, the signals are generated by ______

a) Machine Instructions

b) System programs

c) Utility tools

d) NOTA

Ans: a

2. A word whose individual bits represent a control signal is ______

a) Command word

b) Control word

c) Co-ordination word

d) Generation word

Ans: b

3. A sequence of control words corresponding to a control sequence is called: ______

a) Micro routine

b) Micro function

c) Micro procedure

d) NOTA

Ans: a

4. Individual control words of the micro routine are called as ______

a) Micro task
b) Micro operation
c) Micro instruction
d) Micro command

Ans: c

5. The special memory used to store the micro routines of a computer is ____

a) Control table
b) Control store
c) Control mart
d) control shop

Ans: b

6. To read the control word sequentially ____ is used.

a) PC
b) IP
c) μPC
d) NOTA

Ans: c
Micro program counter

7. Every time a new instruction is loaded into IR, the output of ____ is loaded into μPC.

a) Starting address Register
b) Loader
c) Linker
d) Clock

Ans: a

The starting address generator is used to load the address of the next micro instruction.

1. BIOS is a type of system software, which is stored in ____ located on the motherboard.

1. DRAM
2. SRAM
3. Cache Memory
4. ROM

Ans: 4

2. Suppose a CPU has a clock speed of 2 GHz. How many clock cycles are required to execute an instruction that takes 4 clock cycles to complete?

Clock speed is 2 GHz. ie there are

$2 * 10^9$ cycles pers second.

$\therefore$ 1 cycle last for $\frac{1}{2 * 10^9}$ seconds

we know $1 \sec = 10^9$ nanosecond

ie 1 cycle last for $\frac{1}{2 * 10^9} \times 10^9$ nanosecond

$= \frac{1}{2}$ ns

$\therefore$ 4 cycle lasts for $4 \times \frac{1}{2}$

$= 2\,ns$

3. Which of the following hazards occur at the following 2 instructions in pipeline?

ADD R4, R2, R8.

ADD R4, R7, R4.

a) RAW

b) WAR

c) WAW

d) Both a & b

Ans: d

The second instruction reads the value of R4, which was written by the first instruction. This creates a data dependency and RAW hazard.

The second instruction writes to R4 after the first instruction has read the original value of R4. This creates a WAR hazard

In this case, both instructions write to R4, but the second write happens after a read which doesnot create a WAW hazard in this specific scenario.

RAW hazards occurs when instruction J tries to read data before instruction I writes it.

eg: I : $R2 \leftarrow R1 + R3$

J : $R4 \leftarrow R2 + R3$

WAR hazards occurs when instruction J tries to write data before instruction I reads it.

eg: I : $R2 \leftarrow R1 + R3$

J : $R3 \leftarrow R4 + R5$

WAW hazard occurs when instruction J tries to write output before instruction I writes it

eg: I : $R2 \leftarrow R1 + R3$

J : $R2 \leftarrow R4 + R5$

4. When we perform subtraction on -7 and -5 the answer in 2's complement form is ______

a) 11110 b) 1110 c) 1010 d) 0010

$-7 - -5 = -7+5$

$7 \rightarrow 0111$

1's complement $\Rightarrow$ 1000 +
1

1001 = -7

Ans: b

1001 = -7 +
0101 = 5

1110 = -2

5. Assume a memory access to main memory on a cache "miss" takes 30 ns and a memory access to the cache (on a cache "hit") takes 3 ns. If 80% of the processor's memory requests results in a cache hit, what is the average memory access time?

a. 8.4 ns
b. 9 ns
c 2.2 ns
d 4.4 ns

Ans: b

cache hit time, $T_1 = 3$ ns

cache miss time, $T_2 = 30$ ns

cache Hit, $H_1 = 80\% = \frac{80}{100} = 0.8$.

cache Miss, $M_1 = 20\% = \frac{20}{100} = 0.2$

Average Memory Access Time $= H_1 T_1 + M_1 (T_1 + T_2)$

$= 0.8 \times 3 + 0.2\,(3+30)$

$= 2.4 + 6.6$

$= 9$ ns

6. A processor has an instruction cache with a hit rate of 90% and an access time of 1ns. If the cache miss penalty is 20 ns, what is the average memory access time ?

a) 1.1 ns b) 2 ns c) 2.2 ns d) 3ns

Cache hit, $H_1 = 90\% = .9$

Cache Miss, $M_1 = 10\% = .1$

cache Hit time $= 1$ ns

cache miss time $= 20$ns

Average Memory Access Time $= H_1 \times T_1 + M_1(T_1 + T_2)$

$= .9 \times 1 + .1(1+20)$

$= .9 + 2.1$

$= \underline{\underline{3\,ns}}$

7. A computer system uses a direct-mapped cache with a cache size of 8 KB and a block size of 32 bytes. How many bits are needed for the cache index.

8 KB

Tag	No. of lines in cache	Block/Line offset

$\hookrightarrow = \dfrac{\text{cache size}}{\text{Line size}}$

$= \dfrac{8\,KB}{32\text{ byte}}$

$= \dfrac{2^3 \times 2^{10}}{2^5}$

$= 2^8$

$\log_2(2^8)$

No. of bits needed for cache index $= \underline{\underline{8\text{ bits}}}$

8. The size of the physical address space of a processor is 2^P bytes. The word length is 2^W bytes. The capacity of cache memory is 2^N bytes. The size of each cache block is 2^M words. For a K-way set-associative cache memory, the length (in number of bits) of the tag field is

A) $P-N-\log_2 K$

B) $P-N+\log_2 K$ Ans: B

C) $P-N-M-W-\log_2 K$

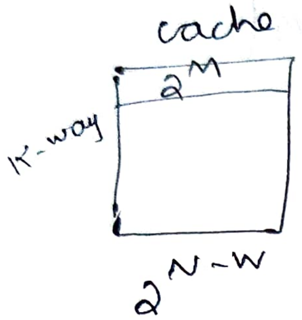
D) $P-N-M-W+\log_2 K$

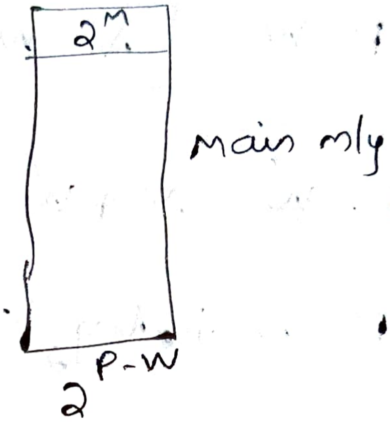
- Physical address space $= 2^P$ bytes
- word length $= 2^W$ bytes
- Capacity of cache $= 2^N$ bytes
- size of cache block $= 2^M$ words
- K-way set associative

physical address space interms of words $= \dfrac{2^P}{2^W}$

$= 2^{P-W}$ words

capacity of cache in words $= \frac{2^N}{2^W} = 2^{N-W}$ words





Physical address space for K-way set associative cache is

$P-W$

Tag	No: of Set	Block/Line offset
		M

$\to \dfrac{\frac{\text{cache Size}}{\text{Line size}}}{\text{Set size}}$

$$= \frac{\frac{2^{N-W}}{2^M}}{K}$$

$\boxed{2^{\log_2 K} = K}$

$$= \frac{2^{N-W-M}}{2^{\log_2 K}}$$

$$= 2^{N-W-M-\log_2 K}$$

ie No: of set $= N-W-M-\log_2 K$

$\therefore$ No: of bits in tag field =

$$P - W - [N - W - \cancel{M} - \log_2 K + \cancel{M}]$$

$$= P - \cancel{W} - N + \cancel{W} + \log_2 K$$

$$= P - N + \log_2 K$$

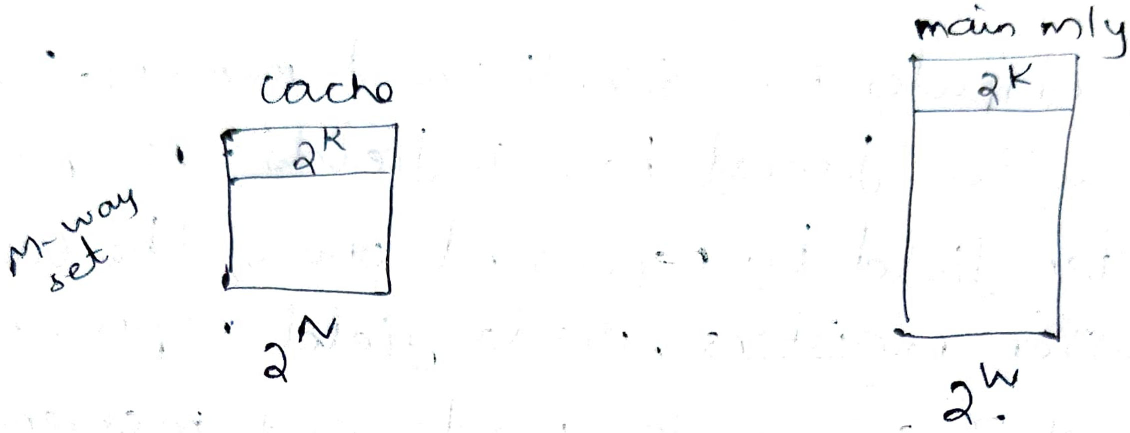
9. The size of the physical address space of a 32-bit processor is 2^W words. The capacity of cache Memory is 2^N words. The size of each cache block is 2^K words. For a M-way set-associative cache memory, the length (in number of bits) of the tag field is

a) $W - N + \log_2 M$ — Ans: a

b) $W - N - \log_2 M$

c) $W - N - K - \log_2 M$

d) $W - N - K + \log_2 M$



physical address space for M-way set associative cache is

W bit

Tag	No: of set	Block/Line offset

$\underbrace{\text{Block/Line offset}}_{K \text{ bit}}$

$\to \dfrac{\dfrac{\text{cache size}}{\text{Line size}}}{\text{set size}}$

$$= \frac{\frac{2^N}{2^K}}{M}$$

$$= \frac{2^{N-K}}{2^{\log_2 M}}$$

$$= 2^{N-K-\log_2 M}$$

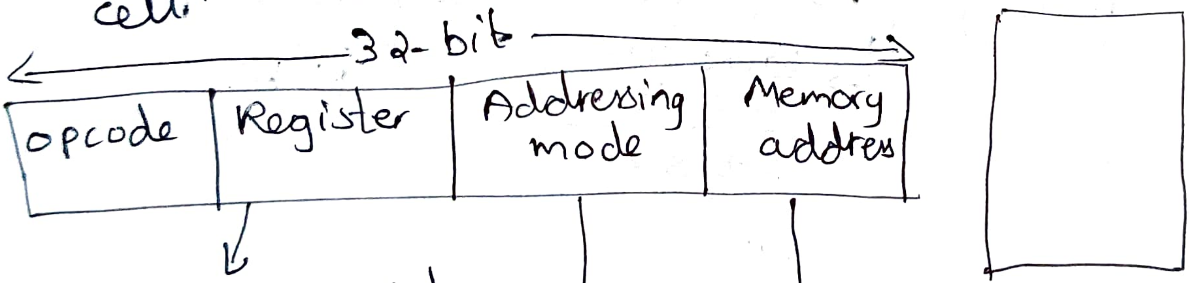
ie No: of set $= N-K-\log_2 M$ bit

$\therefore$ No: of bits in tag field $=$

$$W - [N-K-\log_2 M + K]$$

$$= W - N + \log_2 M$$

10. A computer has 256 K word memory. The instruction format has 4 fields: ie Opcode, register field to represent one of the 60 processor registers, mode field represent one of 7 addressing modes and memory address field. How many instructions the system supports when a 32-bit instruction is placed in the one memory cell.



60 processor register will require 6 bits ($60 < 2^6$)

7 addressing mode will require 3 bits ($7 < 2^3$)

256 K word or Byte

$2^8 \times 2^{10} = 2^{18}$

memory address field will require 18 bits

$\therefore$ opcode $= 32 - (6 + 3 + 18) = 5$

So maximum instructions the system supports

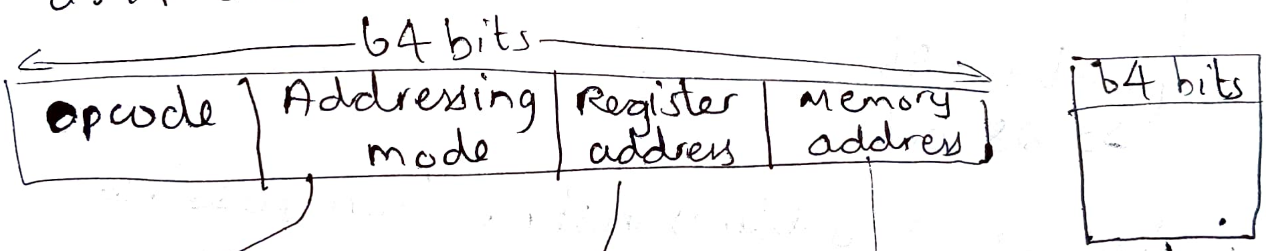
$= 2^5$

$= \underline{32}$

11. The memory unit of a computer has 1 Giga words of 64 bits each. The computer has instruction format, with 4 fields: an opcode field; a mode field to specify one of 12 addressing modes; a register address field to specify one of 48 registers; and a memory address field. If an instruction is 64 bits long, how large is the opcode field?

a) 34 bits   b) 24 bits   c) 20 bits

d) 14 bits



12 addressing mode will require 4 bits ($12 < 2^4$)

48 registers will require 6 bits ($48 < 2^6$)

1 Giga words or bytes

i.e $1 \times 2^{10} \times 2^{10} \times 2^{10}$ byte

i.e 2^{30} bytes

memory address will require 30 bits

$\therefore$ opcode

$= 64 - (4+6+30) = \underline{\underline{24 \text{ bit}}}$

12. A computer has 64 bit instruction and 12 bit address. If there are 250 3-address instruction and 525 2-address instruction, how many 1-address instruction is possible?

Total instruction size = 64 bits

opcode	address 1	address 2	address 3
	12 bit	12 bit	12 bit

$64 - (12+12+12)$

$= 64 - 36$

$= 28$ bit

∴ No: of 3-address instruction possible = 2^{28}

But we have 250 3-address instruction.
So number of opcodes unused and can be used for 2-address instructions

$= 2^{28} - 250$

We know:

No: of 2 address instructions possible = No: of unused or free opcodes from 3 address instruction * $2^{\text{address field size}}$

Hence the number of 2 address instructions possible $= (2^{28} - 250) * 2^{12}$

But we have 525 such instructions as mentioned in the question.

Hence no. of unused opcodes =

$$(2^{28} - 250) * (2^{12} - 525)$$

This can be used for 1 address instruction generation.

So no. of 1-address instructions possible

= No. of unused opcodes in 2-address instruction format. $* 2^{\text{Address field size}}$

$$= \underline{\underline{(2^{28} - 250) * (2^{12} - 525) * 2^{12}}}$$

13. A computer has 64-bit instructions and 28-bit address. Suppose there are 252 two-address instructions. How many 1-address instructions can be formulated?

a) 2^{24} b) 2^{26} c) 2^{28} d) 2^{30}

Total instruction size = 64 bit

opcode	~~instru~~ address 1	address 2
	28 bit	28 bit

$64 - (28+28)$

$= 64 - 56$

$= 8$

$\therefore$ No: of 2-address instructions possible

$= 2^8$

But we have only 252 2-address instruction as mentioned in the question.

So no: of opcodes unused and can be used for 1-address instruction

$= 2^8 - 252$

$\therefore$ No: of 1-address instruction possible

$= (2^8 - 252) * 2^{28}$

$= (256 - 252) * 2^{28}$

$= 4 \times 2^{28}$

$= 2^2 * 2^{28}$

$= \underline{\underline{2^{30}}}$

14. Determine the number of clock cycles required to process 200 tasks in a six-segment pipeline. (Assume there were no stalls), each segment takes 1 cycle.

a) 1200 cycles
c) 207 ~~bytes~~ cycles
b) 206 cycles
d) 205 cycles

Ans

First task would need 6 clock cycle to come out of pipeline and then remaining 199 tasks would need 1 clock cycle each $= 6 + 199$

$= \underline{\underline{205}}$ clock cycles

15. How many 32 K × 1 RAM chips are needed to provide a memory capacity of 256 K-bytes?

Ans: c

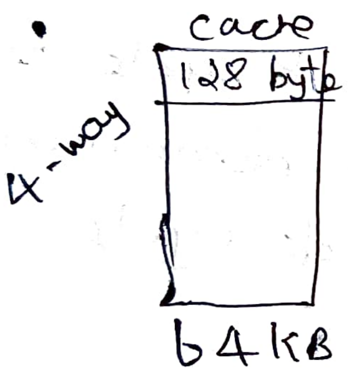
a) 8
b) 32
c) 64
d) 128

32K × 1 means there are 32 K locations each of 1 bit

$$\frac{256\,K \times 1\,\text{byte}}{32\,K \times 1\,\text{bit}} = \frac{256\,K \times 8\,\text{bit}}{32\,K \times 1\,\text{bit}} = 8 \times 8 = \underline{\underline{64}}$$

assume

16. A cache has a 64 KB capacity; 128-byte lines (blocks) and is 4-way set associative. The system containing the cache uses 32-bit addresses. How many sets does the cache have.



$$\text{No. of sets} = \frac{\frac{\text{Cache Size}}{\text{Line Size}}}{\text{set. Size}}$$

$$= \frac{\frac{64\text{ KB}}{128\text{ byte}}}{4}$$

$$= \frac{\frac{2^6 \times 2^{10}}{2^7}}{2^2}$$

$$= \frac{2^9}{2^2} = 2^7 = \underline{\underline{128}}$$